THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these numbers is divisible by 9? 1

(a) 2359 **(b)** 2357 **(c)** 2360 **(d)** 2358

2. In the Collatz rule (halve if even, multiply by 3 and add 1 if odd), the number after 9 is: 1

(a) 27 **(b)** 28 **(c)** 30 **(d)** 29

3. The sum of the first 12 odd numbers is: 1

(a) 146 **(b)** 144 **(c)** 143 **(d)** 145

4. A two-digit number has tens digit 8 and units digit 0. Its value is: 1

(a) 81 **(b)** 80 **(c)** 82 **(d)** 79

5. The product of an even number and any whole number is: 1

(a) a prime number **(b)** even **(c)** odd **(d)** a multiple of 3

6. The sum of the digits of 7254 is: 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 20

7. The reverse of 5684 is: 1

(a) 5684 **(b)** 4865 **(c)** 4568 **(d)** 6845

8. Which of these numbers is a palindrome? 1

(a) 90906 **(b)** 90609 **(c)** 96009 **(d)** 90690

9. Which of these numbers is divisible by 9? 1

(a) 7894 **(b)** 7895 **(c)** 7893 **(d)** 7892

10. In the Collatz rule (halve if even, multiply by 3 and add 1 if odd), the number after 20 is: 1

(a) 12 **(b)** 11 **(c)** 9 **(d)** 10

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Without dividing, test whether 2548 is divisible by 3 and by 9. 2

12. Write the Collatz sequence that starts from 3 (halve if even; triple and add 1 if odd) until you reach 1. 2

13. Use reverse-and-add on 67 until a palindrome appears. 2

14. Without dividing, test whether 3282 is divisible by 3 and by 9. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply one step of Kaprekar's routine to 7502 (largest arrangement of the digits minus smallest arrangement). 3

16. The sum of three consecutive numbers is 87. Find the numbers. 3

17. Without calculating, decide whether $21 \times 3 + 7$ is even or odd, and give a reason. 3

OR

Apply one step of Kaprekar's routine to 9631 (largest arrangement of the digits minus smallest arrangement). 3

18. The sum of three consecutive numbers is 156. Find the numbers. 3

OR

Without calculating, decide whether $14 \times 5 + 10$ is even or odd, and give a reason. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Take the two-digit number 14. Add it to its reverse, show that the sum is divisible by 11 and find the quotient. 5

OR

Write the Collatz sequence starting from 14. Find the number of steps to reach 1, the largest number reached and how many even numbers occur. 5

20. Take the two-digit number 68. Add it to its reverse, show that the sum is divisible by 11 and find the quotient. 5

OR

Write the Collatz sequence starting from 15. Find the number of steps to reach 1, the largest number reached and how many even numbers occur. 5

END OF QUESTION PAPER — Total: 40 Marks

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